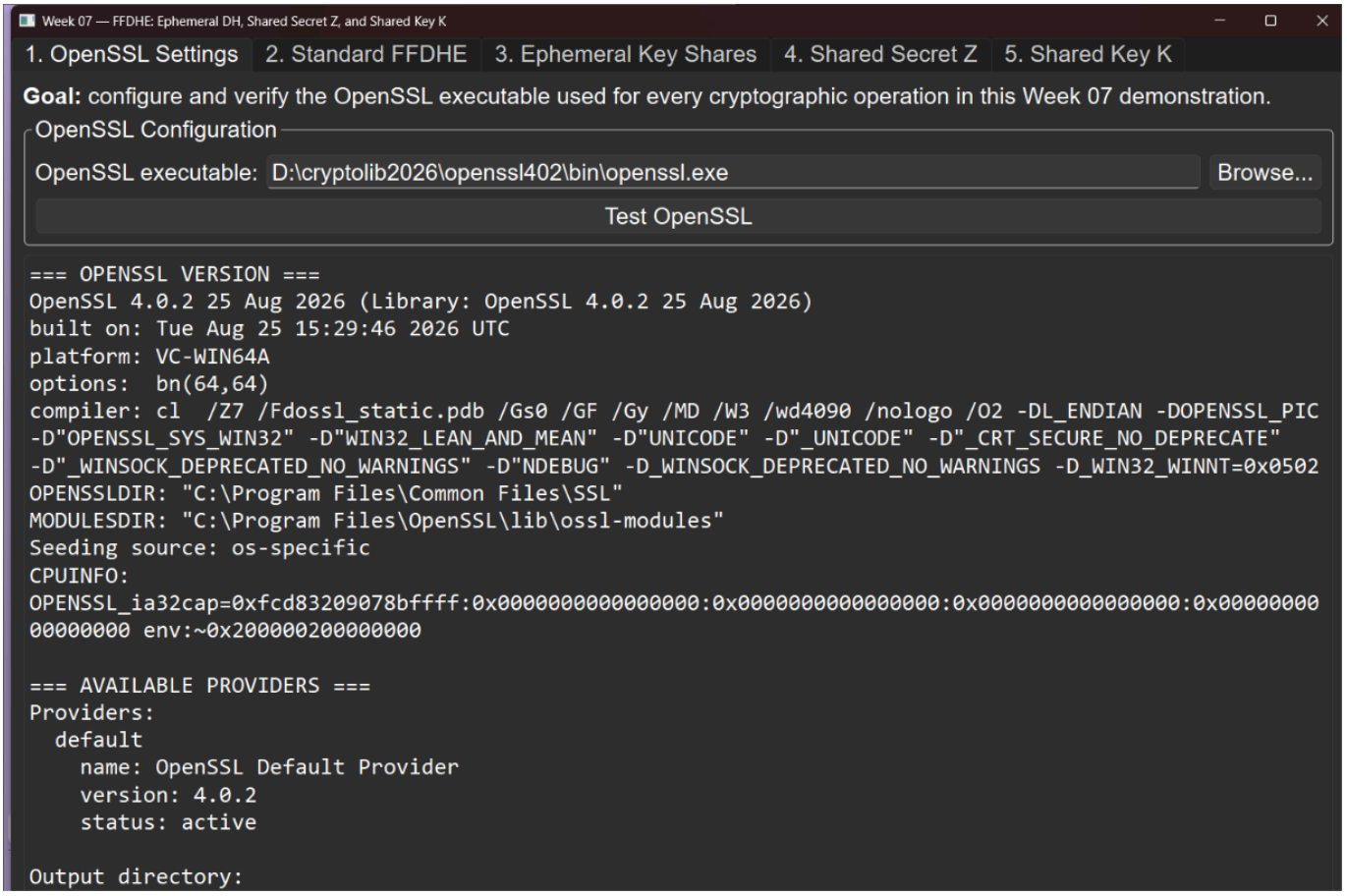
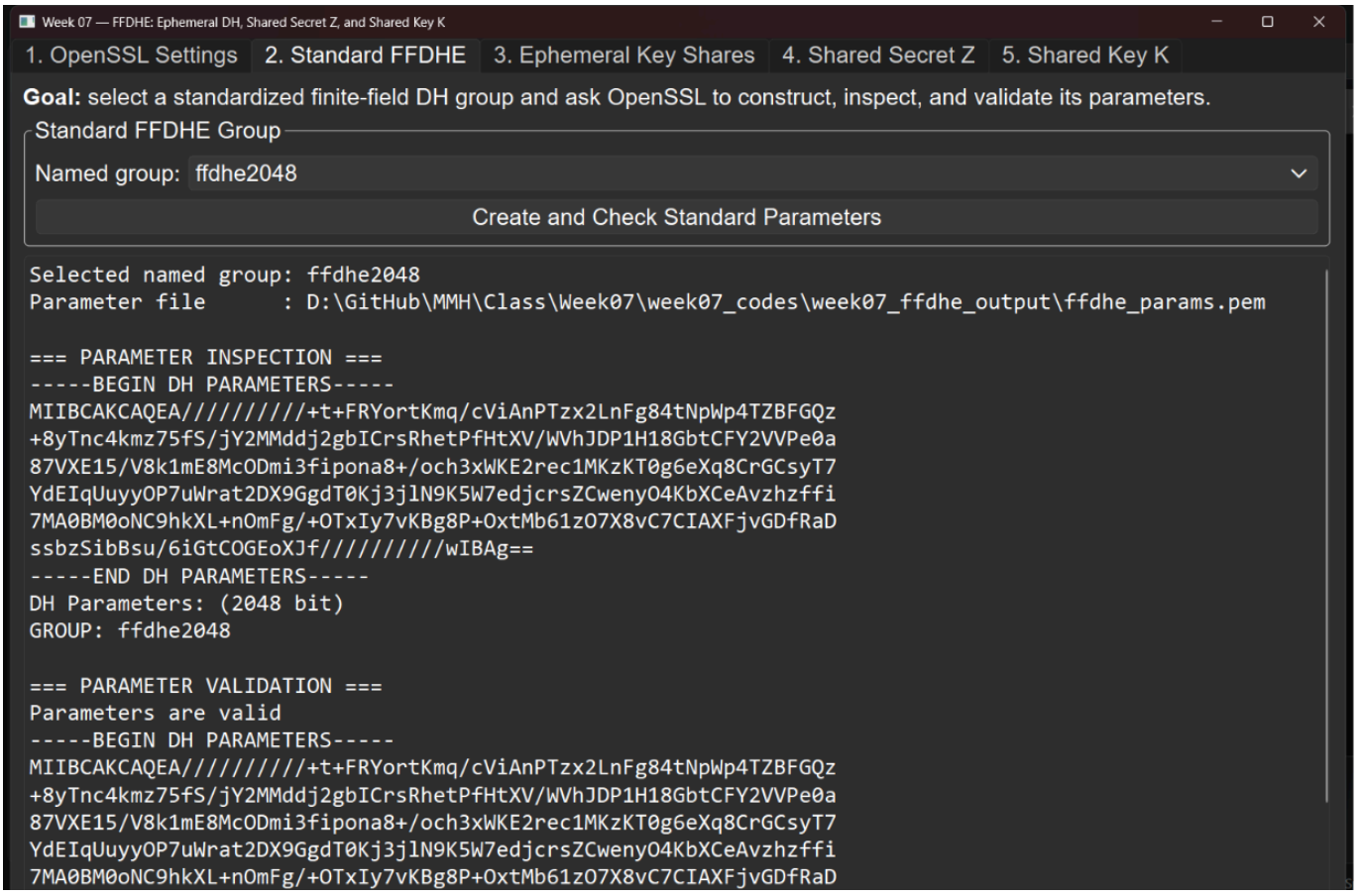
Chọn trường 2048

1. Test openssl

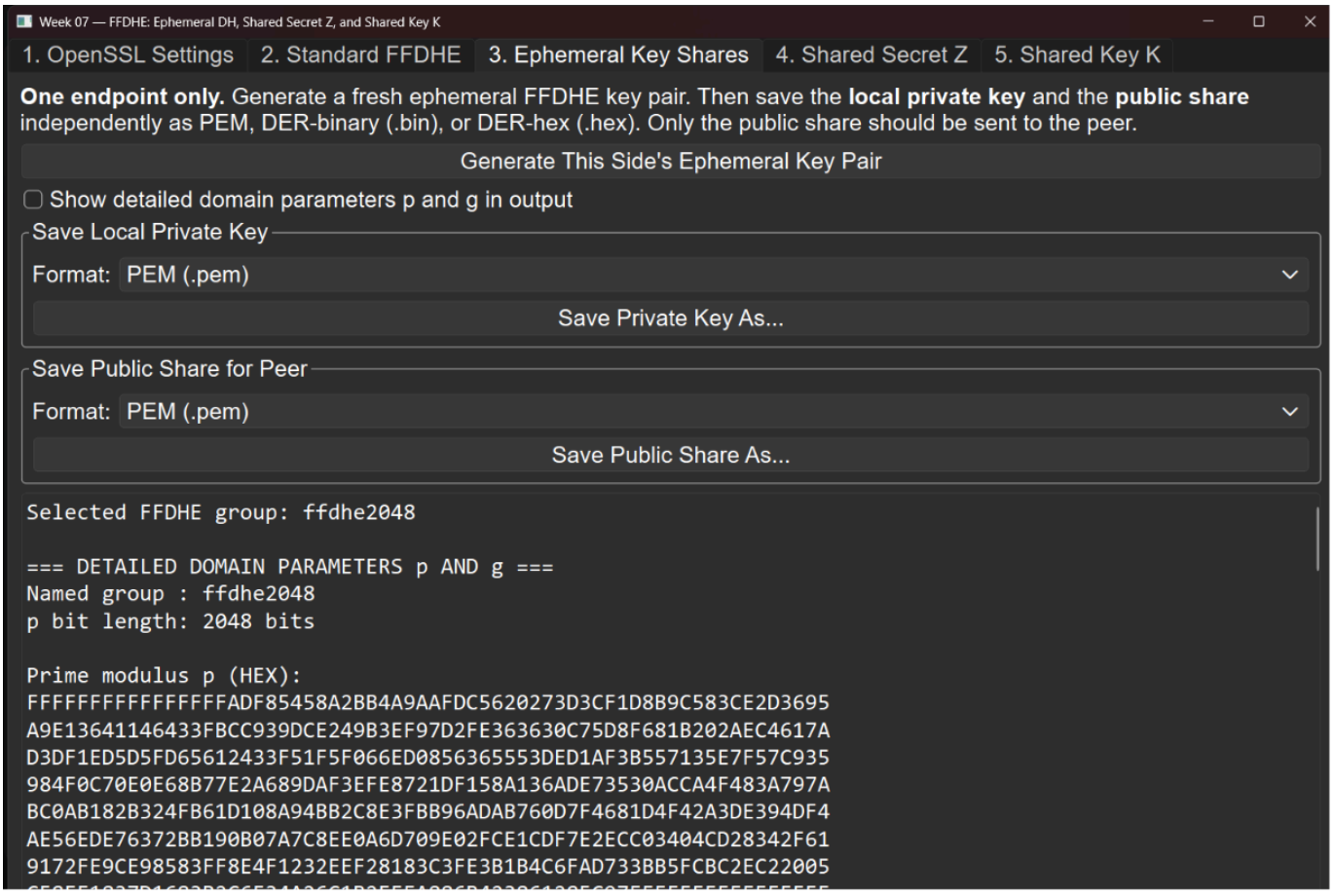


2. Tạo FFHDE chọn trường 2048

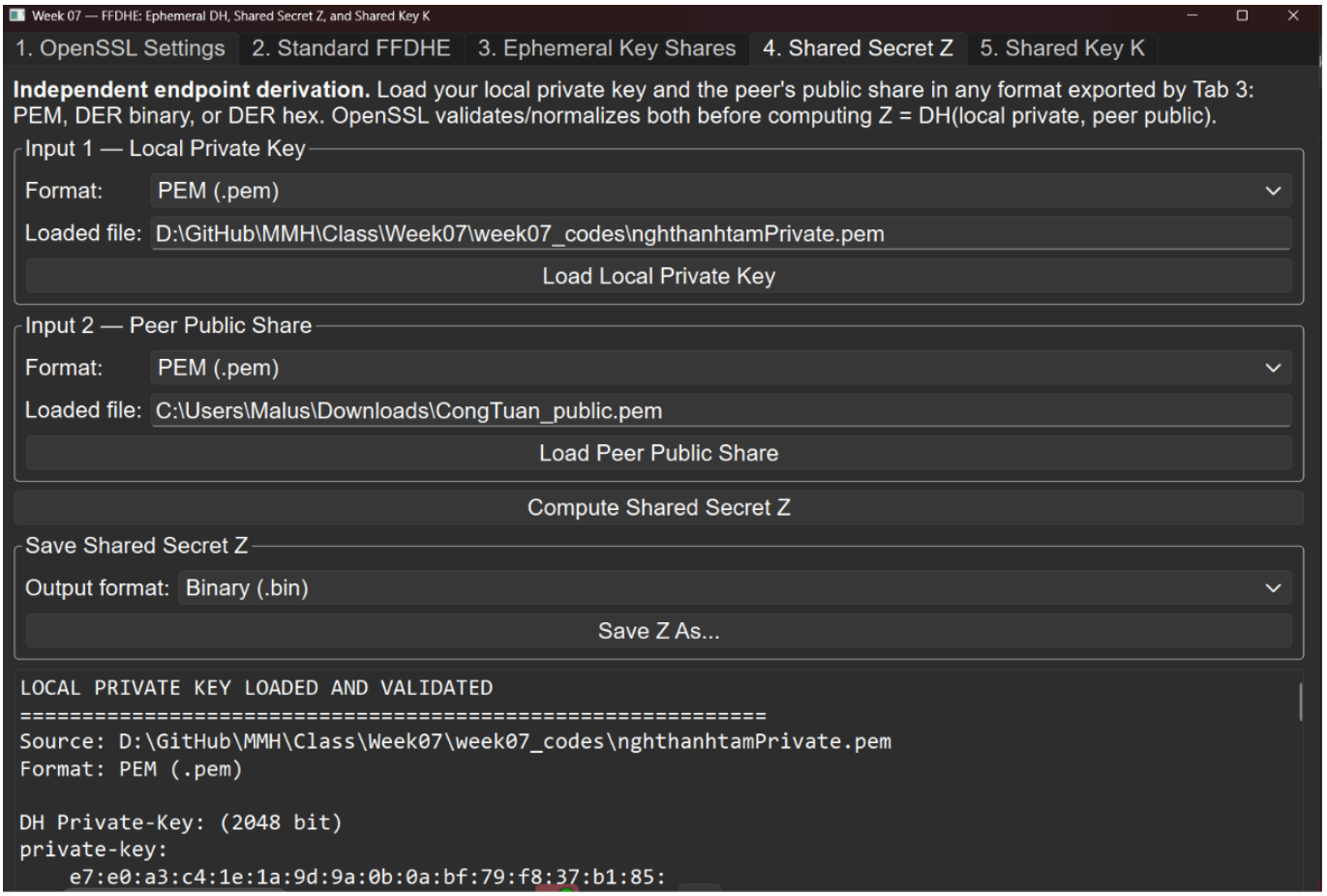


nên gửi file binary gửi public, tính key qua private tính key aes dùng hàm băm 256

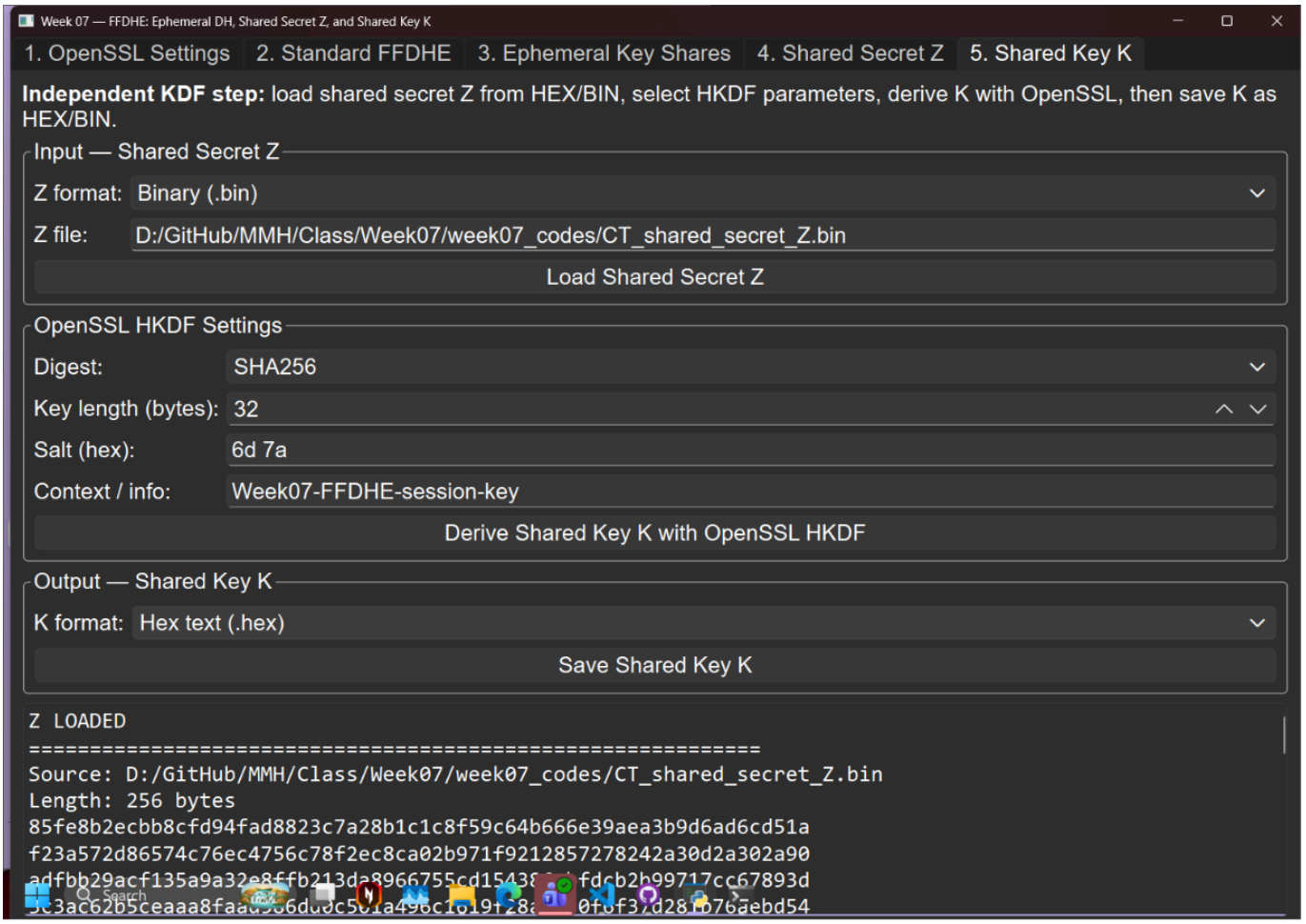
3



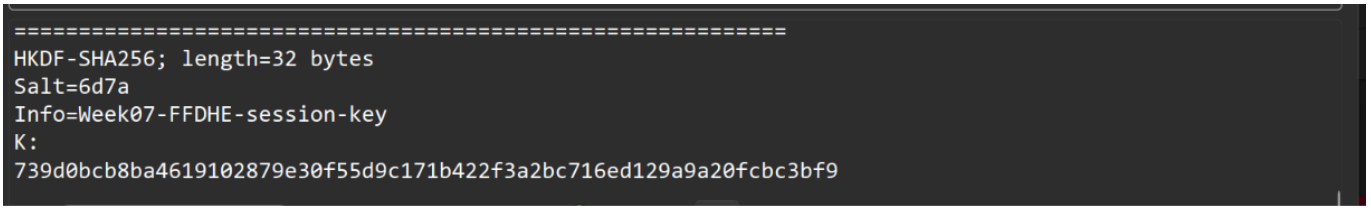
4

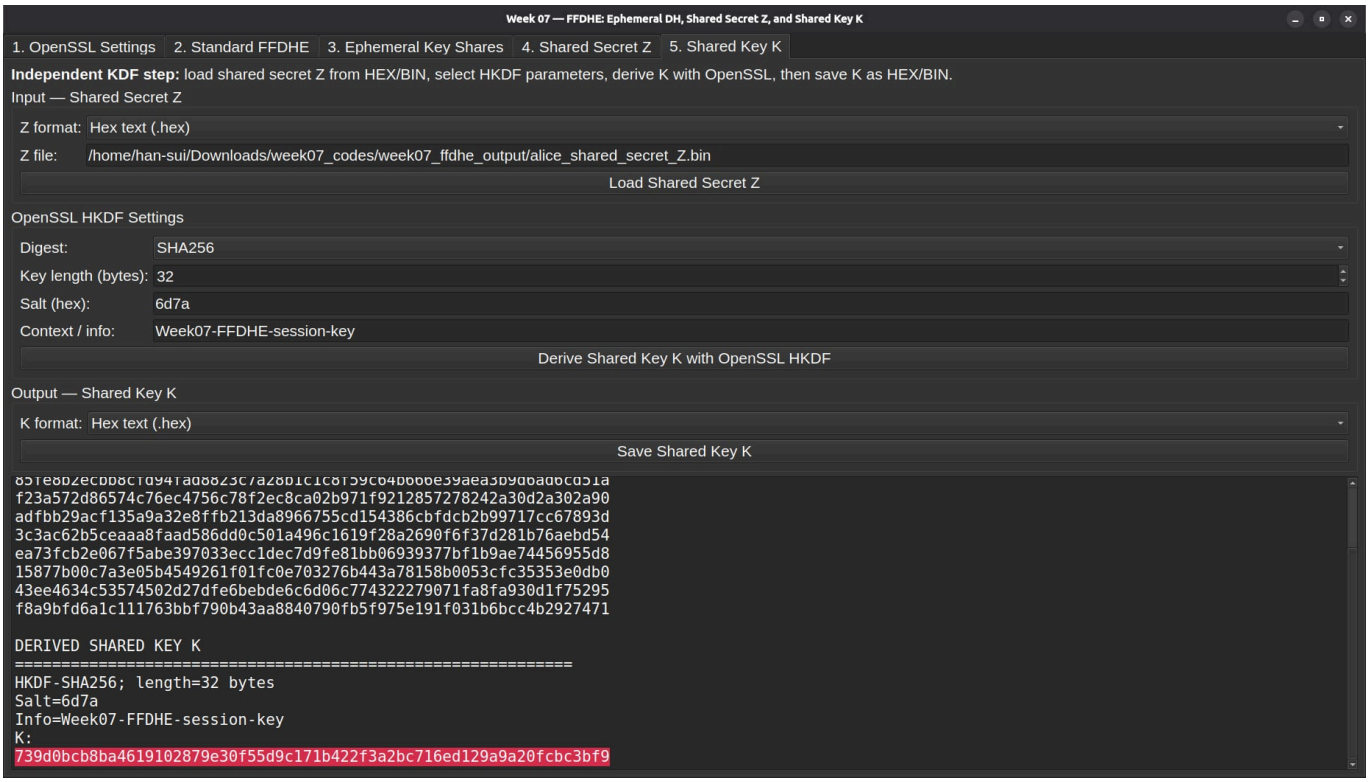

Week 07 — FFDHE: Ephemeral DH, Shared Secret Z, and Shared Key K
1. OpenSSL Settings
2. Standard FFDHE
3. Ephemeral Key Shares
4. Shared Secret Z
5. Shared Key K
Independent endpoint derivation. Load your local private key and the peer's public share in any format exported by Tab 3: PEM, DER binary, or DER hex. OpenSSL validates/normalizes both before computing Z = DH(local private, peer public).
Input 1 — Local Private Key
Format: PEM (.pem)
Loaded file: D:\GitHub\MMH\Class\Week07\week07_codes\nghthanhtamPrivate.pem
Load Local Private Key
Input 2 — Peer Public Share
Format: PEM (.pem)
Loaded file: C:\Users\Malus\Downloads\CongTuan_public.pem
Load Peer Public Share
Compute Shared Secret Z
Save Shared Secret Z
Output format: Binary (.bin)
Save Z As...
LOCAL PRIVATE KEY LOADED AND VALIDATED
Source: D:\GitHub\MMH\Class\Week07\week07_codes\nghthanhtamPrivate.pem
Format: PEM (.pem)
DH Private-Key: (2048 bit)
private-key:
e7:e0:a3:c4:1e:1a:9d:9a:0b:0a:bf:79:f8:37:b1:85:

5


Week 07 — FFDHE: Ephemeral DH, Shared Secret Z, and Shared Key K
1. OpenSSL Settings
2. Standard FFDHE
3. Ephemeral Key Shares
4. Shared Secret Z
5. Shared Key K
Independent KDF step: load shared secret Z from HEX/BIN, select HKDF parameters, derive K with OpenSSL, then save K as HEX/BIN.
Input — Shared Secret Z
Z format: Binary (.bin)
Z file: D:/GitHub/MMH/Class/Week07/week07_codes/CT_shared_secret_Z.bin
Load Shared Secret Z
OpenSSL HKDF Settings
Digest: SHA256
Key length (bytes): 32
Salt (hex): 6d 7a
Context / info: Week07-FFDHE-session-key
Derive Shared Key K with OpenSSL HKDF
Output — Shared Key K
K format: Hex text (.hex)
Save Shared Key K
Z LOADED
Source: D:/GitHub/MMH/Class/Week07/week07_codes/CT_shared_secret_Z.bin
Length: 256 bytes
85fe8b2ecbb8cfd94fad8823c7a28b1c1c8f59c64b666e39aea3b9d6ad6cd51a
f23a572d86574c76ec4756c78f2ec8ca02b971f9212857278242a30d2a302a90

6


==========
HKDF-SHA256; length=32 bytes
Salt=6d7a
Info=Week07-FFDHE-session-key
K:
739d0bcb8ba4619102879e30f55d9c171b422f3a2bc716ed129a9a20fcbc3bf9


Week 07 — FFDHE: Ephemeral DH, Shared Secret Z, and Shared Key K
1. OpenSSL Settings
2. Standard FFDHE
3. Ephemeral Key Shares
4. Shared Secret Z
5. Shared Key K
Independent KDF step: load shared secret Z from HEX/BIN, select HKDF parameters, derive K with OpenSSL, then save K as HEX/BIN.
Input — Shared Secret Z
Z format: Hex text (.hex)
Z file: /home/han-sui/Downloads/week07_codes/week07_ffdhe_output/alice_shared_secret_Z.bin
Load Shared Secret Z
OpenSSL HKDF Settings
Digest: SHA256
Key length (bytes): 32
Salt (hex): 6d7a
Context / info: Week07-FFDHE-session-key
Derive Shared Key K with OpenSSL HKDF
Output — Shared Key K
K format: Hex text (.hex)
Save Shared Key K
f23a572d86574c76ec4756c78f2ec8ca02b971f9212857278242a30d2a302a90
adfbb29acf135a9a32e8ffb213da8966755cd154386cbfdcb2b99717cc67893d
3c3ac62b5ceaaa8faad586dd0c501a496c1619f28a2690f6f37d281b76aebd54
ea73fcb2e067f5abe397033ecc1dec7d9fe81bb06939377bf1b9ae74456955d8
15877b00c7a3e05b4549261f01fc0e703276b443a78158b0053cfc35353e0db0
43ee4634c53574502d27dfe6bebde6c6d06c774322279071fa8fa930d1f75295
f8a9bfd6a1c111763bbf790b43aa8840790fb5f975e191f031b6bcc4b2927471
DERIVED SHARED KEY K
==========
HKDF-SHA256; length=32 bytes
Salt=6d7a
Info=Week07-FFDHE-session-key
K:
739d0bcb8ba4619102879e30f55d9c171b422f3a2bc716ed129a9a20fcbc3bf9